

# Sarah Yuzi Sandström

Computer Science student · Python · SQL · Data pipelines · Testing & CI

Kauniainen, Finland · Permanent Finnish residency · +358 45 102 9420  
sarahyuzisandstrom@gmail.com · LinkedIn · GitHub · <https://yuzi2021.github.io/portfolio-data-ai/>

## PROFILE

Computer Science student (University of London, expected 2027) who has been building software through my studies and independent projects since 2023, after an established career in education. My recent work covers Python, SQL and PostgreSQL, relational data modelling, REST APIs, validated ETL, automated testing, Git, GitHub Actions, Docker and deployment. I am working towards data and AI engineering and want to develop this foundation in a professional engineering environment.

## TECHNICAL SKILLS

|                        |                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming & data     | Python · SQL · PostgreSQL · SQLite · relational data modelling · data validation & transformation                                                        |
| Backend & applications | Django · Django REST Framework · REST APIs · OpenAPI docs (Swagger, ReDoc) · Streamlit · JavaScript · HTML/CSS                                           |
| Development & quality  | Git · GitHub · GitHub Actions · CI/CD · automated testing (Django test runner, unittest, pytest) · Docker · Docker Compose · Render                      |
| Data & AI              | ETL (extract, validate, transform, load) · scheduled ingestion & change detection · TensorFlow/Keras · scikit-learn · SHAP · NumPy · pandas · Claude API |

## PROJECTS

**SkillPath Navigator — data pipeline & REST API** Coursework, extended independently · 2026  
[Live demo](#) · [source repository](#) [private](#)

- Django REST Framework API over 464 job postings, 319 skills and 5,845 job–skill relationships, with a many-to-many model through an explicit join table.
- Built an extract → validate → transform → load command: malformed rows, conflicting duplicates and broken cross-file references reject the snapshot before an atomic, repeatable load.
- 44 passing tests, run by GitHub Actions against PostgreSQL 16; Docker Compose for development; SQLite demo deployed on Render only after checks pass.

**Phishing email classifier: signal or artifact?** CM3015 Machine Learning coursework · 2026

- Parsed the Nazario (mbox) and Enron (maildir) corpora into one record format; deduplicated before splitting, balanced and stratified the data; fitted the vocabulary on training data only.
- TensorFlow Dense/Dropout models compared across three seeds; test F1 0.970–0.998. SHAP showed 18 of the 25 most influential tokens in the headers model were source artifacts, and recall fell to 0.38 on other benchmark corpora.

**GRC Change Monitor — scheduled ingestion & change detection** Self-directed · 2026  
[Live demo](#) · [source repository](#) [private](#)

- Weekday GitHub Actions pipeline reading six official sources via RSS, JSON and HTML adapters; normalises and fingerprints items and stores changes, history and source health in SQLite.
- Partial failures never report as “no changes”; first-run items recorded as a baseline; 67 deterministic tests run before each monitoring run.

**Also:** SMS spam classification with Naïve Bayes implemented from scratch in NumPy (coursework) · [AI research digest agent](#) (GitHub Actions, Claude API) · GRC Ops Lab (CSV reconciliation, 35 pytest tests)

## PROFESSIONAL EXPERIENCE

**Privacy & Digital-Workplace Development Project** 2024–2026  

- Investigated digital-tool and information-handling problems with staff, management, IT and a Data Protection Officer; produced a legally reviewed 57-page staff guide and training. Published as a co-authored thesis (Laurea, 2024).

**Teacher Assistant — Pilke Päiväkodit, Finland** 2022–2024  

- Resolved issues independently in a fast-moving service environment; designed and delivered staff learning on data protection, information security and responsible use of digital tools.

**Educator / Pedagogy Consultant — International schools & education technology** 2013–2022  

- At LingoAce, automated the reporting system, reducing reporting time by 70%. Developed learning materials and worked across multicultural teams, adapting explanations to each audience.

## EDUCATION

**BSc Computer Science — University of London** Expected 2027  
Web development · databases · APIs · software testing · machine learning

Master’s studies in Changing Education, University of Helsinki (2025) · MA English Studies & Subject Teacher Training, University of Helsinki (2019) · BA English Studies for Teaching, University of Buckingham, First Class Honours (2013)  
**Languages:** Mandarin (native) · English (full professional) · Finnish (basic to pre-intermediate) · Swedish (elementary)